

Muscle Cramps

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A cramp is a sudden, brief, unintended (involuntary), and usually painful contraction of a muscle or group of muscles. Muscle cramps are common and often benign but occasionally can be a [symptom of nervous system malfunction](#).

Causes of Muscle Cramps

The **most common causes** of muscle cramps are:

- Benign leg cramps that occur for no known reason, typically at night
- Exercise-associated muscle cramping (cramping during or immediately after exercise)

Muscle cramps (also called charley horses) often occur in healthy people, usually in middle-aged and older adults but sometimes in younger people. Cramps tend to occur during or after vigorous exercise but sometimes occur during rest. Some people have painful leg cramps during sleep. [Sleep-related leg cramps](#) usually affect the calf and foot muscles, causing the foot and toes to curl downward. Although painful, these cramps are usually not serious and are thus called benign leg cramps.

Almost everyone has muscle cramps at some time, but certain conditions increase the risk and/or severity of cramps. They include the following:

- Having tight calf muscles, which may be caused by not stretching, inactivity, or sometimes repeated accumulation of fluid (called edema) in the lower leg
- Becoming [dehydrated](#)
- Having low levels of electrolytes (such as [potassium](#), [magnesium](#), or [calcium](#)) in the blood

- Having a nerve disorder or an underactive thyroid gland ([hypothyroidism](#))
- Taking certain medications

Low electrolyte levels may result from use of some diuretics, an alcohol use disorder certain hormonal (endocrine) disorders, [vitamin D deficiency](#), or conditions that cause loss of fluids (and thus electrolytes). Electrolyte levels may become low late in pregnancy.

Cramps can occur shortly after [dialysis](#), possibly because dialysis removes too much fluid from the body, removes the fluid too quickly, and/or lowers electrolyte levels.



Conditions That Cause or Contribute to Muscle Cramps

Category	Examples
Medications and other substances	
Use of certain medications and other substances	Angiotensin II receptor blockers and some beta-blockers (used to treat high blood pressure), bronchodilators (used to treat asthma), <u>cisplatin</u>, diuretics, <u>donepezil</u>, <u>lovastatin</u>, birth control pills (oral contraceptives), <u>pyrazinamide</u>, <u>raloxifene</u>, synthetic <u>parathyroid hormone</u> (<u>teriparatide</u>), <u>tolcapone</u>, or <u>vincristine</u> Stimulants, such as amphetamines, caffeine, cocaine, <u>ephedrine</u>, nicotine, or <u>pseudoephedrine</u>
Sudden stopping of a drug or medication	Sedatives, such as alcohol, barbiturates, or benzodiazepines Medications used to treat insomnia or anxiety
Disorders	
Metabolic and hormonal (endocrine) disorders	Low levels of potassium (<u>hypokalemia</u>), low levels of magnesium (<u>hypomagnesemia</u>), or low levels of calcium (<u>hypocalcemia</u>) <u>Alcohol use disorder</u> <u>Hypothyroidism</u> (an underactive thyroid gland)
Musculoskeletal disorders	Tight calf muscles



Metabolic and hormonal
(endocrine) disorders

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